

THE UNIVERSITY OF DANANG
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**Gene expression profiling of THP-1 cells treated
with 4,15-iso-atriplicolide tiglate by two
different RNA-seq analysis pipelines**

by

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Da Nang , November 2021

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DECLARATION

I, Nguyễn Đình An from BMS17, hereby declare that the present diploma thesis [Gene expression profiling of THP-1 cells treated with 4,15-iso-atricolicide tiglate by two different RNA-seq analysis pipelines] was composed by myself and that the work contained herein is my own. I confirm that:

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Nguyễn Đình An

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ABBREVIATIONS

Abbreviations	Name
DNA	Deoxyribonucleic Acid
RNA-seq	RNA Sequencing
FDR	False discovery rate
DE	Different expression
AT	Atriplicolide tiglate
ADCC	Antibody-dependent cell-mediated cytotoxicity
MDSC	Myeloid-derived suppressor cells

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ABSTRACT

Several recent advancements in RNA-seq techniques have allowed for a more comprehensive characterization of RNA transcripts. Continued advancements in RNA-seq applications, notably direct RNA sequencing and methods that allow RNA quantifying from extremely tiny quantities of biological materials. With the great development in the way RNA-Seq works, from studying and adding sample data to the library that provides insight into transcriptional information of the cell. The purpose of this study was to analyze 4,15-iso-atriplicolide tiglate (AT) a novel inhibitor of MYB-dependent transcription has therapeutic potential against AML. Using RNA sequencing analysis for transcript recognition and gene expression quantification across two pipelines, each using different tools and follow these main steps include quality control of the raw data, pre-processing step, read the alignment, transcript assembly and qualification after that is differential gene expression. It was found that a total of nearly 6,300 genes, including 2900 up-regulated genes and 3400 down-regulated genes, were significantly differentially expressed in AT-affected cells. In addition, functional annotation analysis against the Gene Ontology database revealed that the target genes were differentially expressed and significantly enriched in , innate immune response, microglial cell activate reaction, neutrophil degranulation, response to virus. Thereby expanding the database of monocyte cell line derived from acute mononucleosis THP-1, and a comprehensive view of the new potential regulator for studies, which has help prevent and treat acute myeloid leukemia in the future.

Keyword: RNA-seq , 4,15-iso-atriplicolide tiglate, differential expression, gene expression, Gene Ontology.

CHAPTER 1. INTRODUCTION

1.1 RNA Sequencing

Transcriptome is the set of RNA molecules, including messenger RNA, transfer RNA ribosomal RNA and other noncoding RNA molecules that are present in cells. The transcriptome not only reveals aspects of the function of the genome, but also about the different types of biomolecules present in cells or tissues. Microarray methods are based on nucleic acid hybridization and are often used in development of gene expression in the early years of transcriptome research. However, there are access restrictions and large funds, and there are many error-prone steps in microarray experiments, so the experiment needs to be repeated to eliminate the source causing an error, each microarray can only be included on the array. Because hybridization focuses on sequenced genes and recognized genomic areas, another approach has been suggested to solve the microarray methodology. Due to the complexity of sequencing the whole cDNA segment and the cost involved in generating ESTs, many tag-based high-throughput sequencing methods were developed to quantify expression. The high throughput of Next Generation Sequencing (NGS) technology makes the use of RNA-seq for transcriptome analysis possible and economical. RNA-seq extracts RNA from the sample, then selects a subset of the RNA population as poly A tail, then converts it into cDNA, and connects the cDNA to a specific adapter at one or both ends of fragments. These modified fragments are sequenced from single ended or paired end sequencing, with or without amplification. The raw data after preprocess are mapped and quantified according to the genome of a reference(1). The de novo transcriptome would perform (2) in case no reference genome is available. Capability of RNA-seq research to detect splice junctions with a single base resolution, and single-base resolution is very useful for discovering novel genes and transcript (3). However, it can be prone to erroneous interpretations due to the presence of DNA artifacts, this kind of issue can be solved using RNA-seq replicates .

1.2 RNA-Seq Data Analysis Workflow

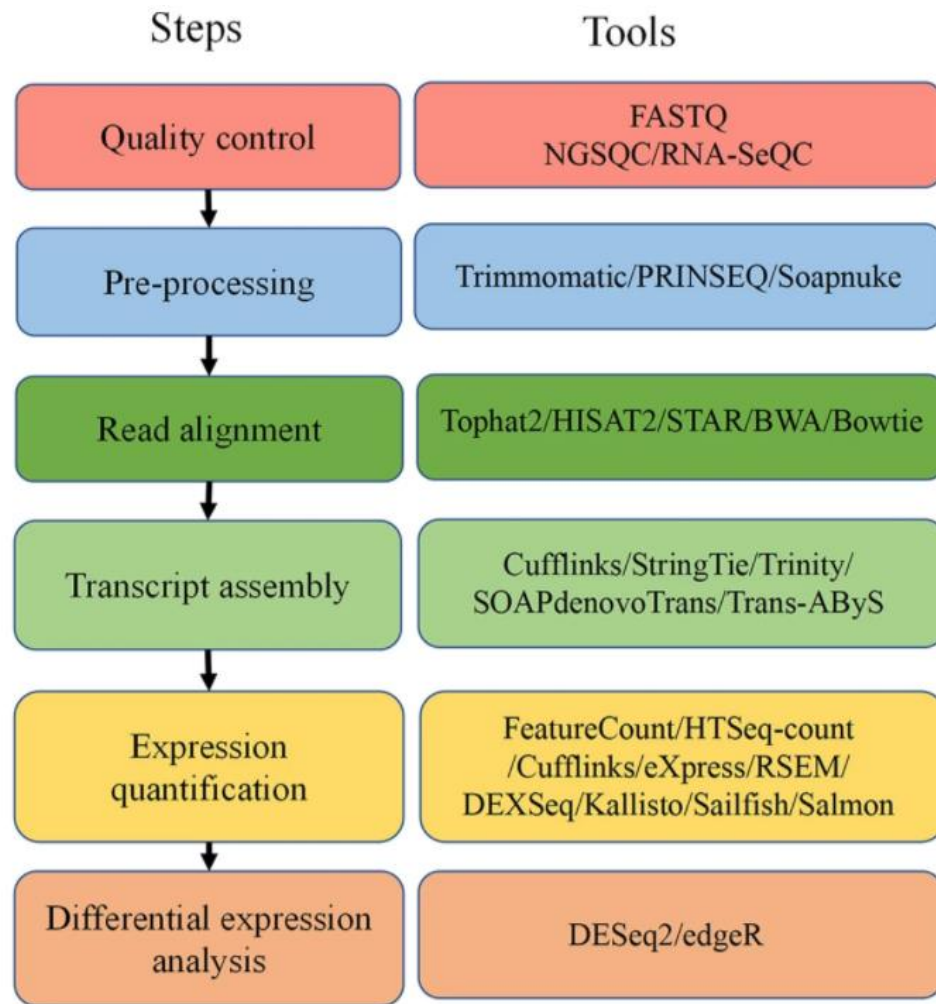


Figure 1: RNA-seq workflow

By using high-throughput RNA-seq, a single study can create large amounts of data, which are assembled correctly for useful interpretation. A significant number of computation methods and tools have been devised regarding efficient data assembly, quantification and differential analysis of such data. There are many popular tools and programs, such as Bowtie, Tophat, and Cufflinks. When algorithms are used to analyze large amounts of RNA-seq data, new isoforms and splice variants as well as unannotated transcripts have been discovered in well studied biological models (8). In general, RNA-seq analysis workflow consists of several steps . As shown in the figure, these steps include quality control of the raw data, pre-processing step, read the alignment, transcript assembly, expression qualification, differential gene expression and pathway analysis.

1.2.1 Quality Control and Pre-processing

Sequencing RNA includes multiple –step processes and is much more complicated. Therefore, evaluating the quality of the sequence is a crucial step before data analysis and RNAseq data can be also preprocessed to meet the expectations of RNA-seq analysis process. Primary steps include data filtering, trimming poor quality sequences, base adapters, overexpressed sequences, and contamination to ensure consistent end results. The quality of the collected raw data will be verified by software such as FastQC (4) or MultiQC to assess the cleanliness of the data; otherwise, the Trimming software will detect them. Development tool for detecting and removing adapters and scanning each reading with a 4-base sliding window, and trimming the lowest-scoring bases and low-quality N bases to improve reads quality.

1.2.2 Read Alignment

After quality control, the initial step in the RNA-seq analysis is to compare the sequencing reads with a reference genome or a transcriptome database. The complexity of mapping RNA-seq reads makes it harder to work with than DNA sequencing reads due to many reads map overlap splice junctions. Indeed, RNA-seq reads using mapping algorithms, such as Bowtie (5) and BWA, are not regularly mapped to the reference genome because they cannot process spliced transcripts. Inserting the sequence generated from the exon-exon splice junction obtained from the known genetic annotation into the reference genome is a technique to solve this problem (1). Use the connector with a signal aligner to map the readings. This method can detect read changes with short insertions when aligned across exon and intron boundaries. Therefore, as the use of RNA-seq data increases, many splicing mapping tools have been developed, especially for mapping transcriptome data. The tools most commonly used to align RNA-seq are: GSNAP, MapSplice, RUM, STAR, HISAT2(6), and TopHat(7). Concerning performance, capability, quickness, and storage capacity, each

aligner offers its own set of benefits. The appropriate aligner to utilize is determined by these criteria as well as the general goals of the RNA-Seq research.

1.2.3 Transcript Assembly and Quantification

The mapping reads could be reassembled into transcripts after RNA-Seq reads are aligned. Most computational algorithms use read alignments with reference genomes to derive transcription models (8–11). De novo reconstruction, which does not require reference to the genome or annotations to construct contiguous transcription sequences, is another method of transcription assembly(12–14). Reconstructing transcripts from short data is a great difficulty, as there is no greater method for assembling transcripts. The nature of the transcriptome, common technical challenges, characteristics of bioinformatics workflows may affect the quality of transcriptome assembly. The estimation of gene expression level is a common downstream function of transcription reconstruction software. Cufflinks (8), FluxCapacitor, FeatureCount (15) and MISO is a calculation tools that counts the number of reads assigned to the complete transcript to quantify. By measuring the quantity of reads that align to an exon, alternative techniques such as HTSeq can measure expression without constructing transcripts (16). On the other hand, this method is not ideal for genes without distinct exons, Cufflinks (17) and MISO use another technique in which researchers construct a likelihood function to simulate sequencing experiments and estimate the highest probability that a read map to a specific isoform.

1.2.4 Differential Expression

The goal of gene expression experiments is to identify transcripts and illustrate differences in expression under different environments. A significant utilization of RNA-seq is the examination of transcriptomes across various formative stages, across an illness state contrasted with ordinary cells, or explicit test boosts contrasted with physiologic circumstances. To perform the analysis requires genetic similarity in order to accurately assess abundance by comparing samples.

From there, it is possible to analyze the functional factors of the genes, the biological mechanisms of the genes, and the mechanisms of disease development. In the early 20th century, new statistical methods were developed to test differential expression and microarray expression, where the normal distribution can be used to estimate the constant intensity of the probe in replicates. Arguably this approach can also be applied to RNA-seq data, but it is limited by different statistical models, which leads to considering readings that do not coincide with the normal distribution. Initial RNA-seq studies showed that the Poisson distribution was consistent with the distribution of reading counts across copies. However several studies showed that the Poisson assumption failed to capture biological changes, leading to a high rate of false positives because of underestimation of sampling error (18). The difference in the reading depth of the artificially generated sequence between samples can be the difference in the original read count between the comparative genes. A more convenient alternative is to convert raw read count data to FPKM or RPKM values for differential expression analysis. In order to find differential expression, several tools were developed, such as a tool Cuffdiff, DESeq and EdgeR, which provide meaning for differentially expressed transcripts to correctly interpret biological observations. In addition to the need to measure variability and estimate model parameters for RNASeq clones, biological replicates are used the necessary number of repetitions for each sample is usually three (19). In some cases, RNA-seq libraries are multiplexed, so additional biological replicates can be used to save sequencing costs and improve the best experimental design.

1.2.5 Pathway Analysis

Genomic data analysis examines the function of genes and their role in gene expression pathways because genes work together in all of their biological processes. Even small changes in gene expression can participate in the pathway and cause significant biological changes. Therefore, the information of the differentially expressed biological sub-pathways provides important explanatory

results (20). Popular analyzes include Gene Ontology and KEGG Pathway Enrichment Analysis. Gene sets are analyzed for enrichment, an enrichment study, for example, can find which GO terms are over-represented using annotations for a group of genes that are up-regulated under certain conditions. KEGG is the process of mapping biological objects such as genes, proteins, small molecules, into molecular interaction networks. This is not only an enrichment process, on the contrary, it is a collection operation that generates a new collection, but also assists in the integration and interpretation of data sets, especially generating a large number of data sets by high-performance technologies. KEGG contains three basic mapping operations: pathway mapping, brite mapping, module mapping. TopGo is a semi-automated enrichment analysis tool that uses Gene Ontology (GO) language. The results of standardized gene expression measurements, gene correlation, or analysis are taken into account in this process. TopGO has a single gene set testing framework, which is one of its key features.(21)

1.3 Applications of RNA Sequencing

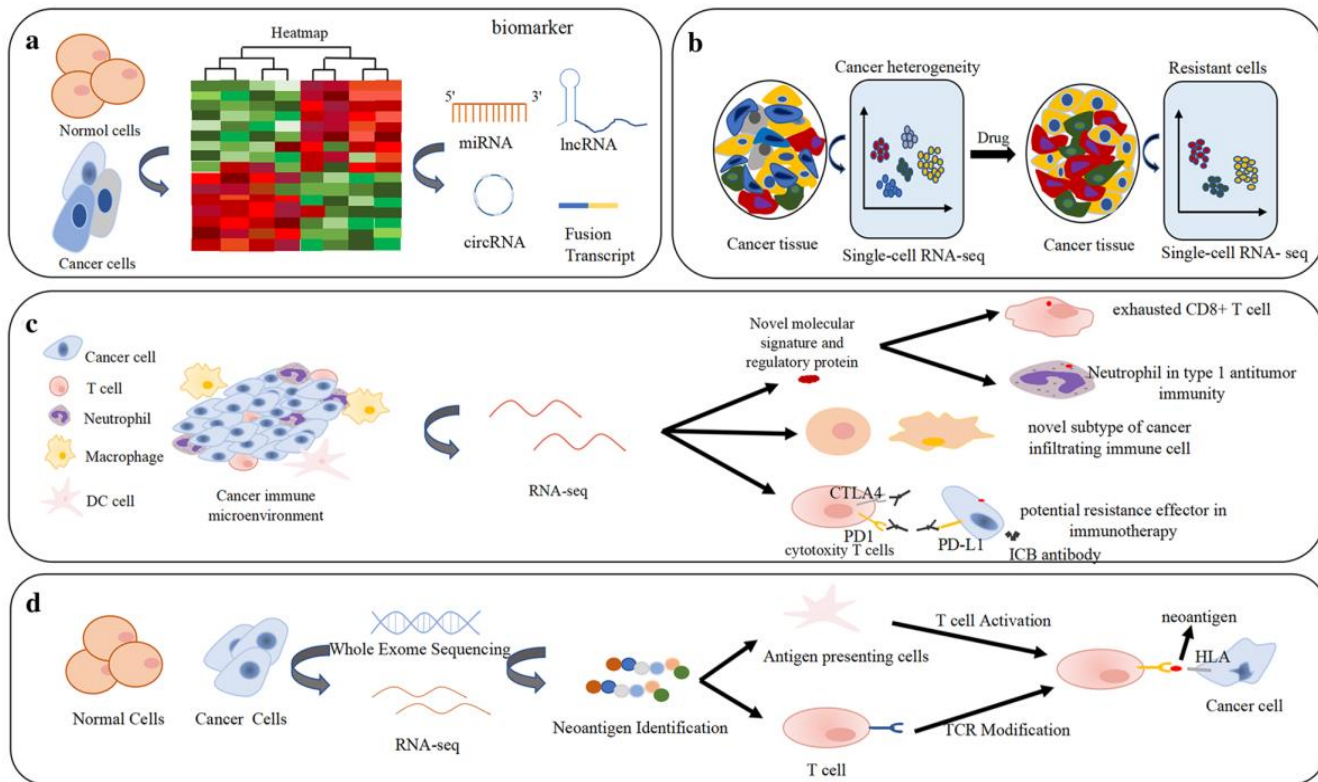


Figure 2: Several ways for application of RNA-seq.

Recent research shows that the important application of RNA sequencing has become increasingly popular over the past several decades. High-throughput sequencing technology (22) has been used extensively, including the analysis of genomic data for cancer research and, more specifically, the analysis of the differential expression of genes, biomarkers in cancer, cancer antibiotics, microenvironment and immunology. The mechanism of drug resistance in cancer treatment is one of the most challenging causes of cancer treatment. In addition, many mechanisms of drug resistance in cancer have not been identified. A study identified the COX7B gene involved in platinum resistance and replaced the CD63 monoclonal antibody in cancer cells with single-cell RNA-seq (23). In a study of bone marrow cells that suppress immune pathways, infiltrate breast cancer tumors, and cause drug resistance, RNA-seq was used to identify genes in bone marrow cells (24). Another common application of RNA sequencing is to analyze the differential expression of genes and biomarkers in cancer, the differential analysis of

gene expression will be a prerequisite for facilitating the discovery of possible biomarkers in cancer (25). Based on the data extraction and analysis of RNA sequencing data and other clinical data, peroxidase isotype is also expected to be a predictive biomarker for predicting breast cancer survival and recurrence-free survival (26) . RNA sequencing is identifying more and more differentially expressed genes, and new potential cancer biomarkers are constantly being discovered. Some findings have also been made on the distinction between benign and malignant tumor cells. The analysis will use SCRNA-seq to find the variations in copy number variations in tumor cells.

1.4 Aim of study

Acute leukemia is a condition of bone marrow malignancy, also known as acute blood cancer, mainly caused by mutations that lead to uncontrolled proliferation and differentiation of white blood cells in the marrow. According to WHO statistics in 2018, the incidence of acute leukemia is 3.9% of the total number of most common cancer cases and the mortality rate is over 65% total(27). Recent studies have shown that transcription factor deregulation of the MYB genes contributes to the development of leukemia and the discovery of a natural compound named 4,15-iso-atrpicolide tiglate (AT) is a novel inhibitor of MYB-dependent transcription has therapeutic potential against acute myeloid leukemia (AML). Several previous studies have shown that MYB-dependent is a proto-oncogene expressed in hematopoietic progenitor cells. The transcription factor Myb regulates blood cell differentiation, proliferation and survival, in addition to being implicated in development and human leukaemic diseases including cancer. The goal of the study will be to analyze RNA-seq data to reveal the effect of AT on gene expression levels, using two different pipelines to compare the differences between these pipeline.

CHAPTER 2. MATERIALS AND METHODS

2.1 Materials

2.1.1 System Information

CPU : AMD Ryzen Threadripper 2920X 12-Core

GPU : GP102 GeForce GTX 1080 TI

Memory : 128 GB

Motherboards : ASUS PRIME X399-A

2.1.2 RNA Sequencing Data

The raw RNA-sequencing data were retrieved from Genome Expression Omnibus database of the National Center for Biotechnology Information (NCBI) with the series accession number GSE128570. Total runs with six samples (SRX5546510, SRX5546511, SRX5546512, SRX5546513, SRX5546514, SRX5546515)

2.2 Methodology

2.2.1 Raw data quality control

Quality control is the process through which a tool renders a report visually, based on the fastqc method. Several ways to encode quality scores affect the performance of the LC-MS system, the Q value is greater than the threshold of 20 to be a good quality, the fundamental read processing processes of cutting base, cutting read, and cutting adapters can be skipped. With a bad sequence the quality value of less than 20 are found, it's probably had to trimdata back to assure the base calls that were received were good.

2.2.2 Pipeline 1 : TopHat, Cufflinks, Cuffdiff

2.2.2.1 Read alignment using TopHat

RNA sequence reads, after being tested and processed, will be aligned to the human reference genome (hg38/GRCh38), by TopHat. TopHat cuts the input reads into smaller segments of actual length and maps these segments independently on the reference genome. After the reading passage is aligned, it will show the result in percentage of mapped read. The uniformity of read coverage on exons and the mapped strand are two more significant criteria. If readings in poly(A)-selected samples predominantly accumulate at the 3' end of transcripts, this could also suggest poor RNA fidelity in the starting material. PCR biases may be revealed by

the GC content of mapped reads. Picard, RSeQC, and Qualimap are tools used for quality control in mapping.

2.2.2.2 Assemble transcripts with Cufflinks

After aligning by TopHat, alignment files were used to create transcriptome assemblies using Cufflinks and create a transcriptome assembly for each condition. These assemblies are then merged together using cuffmerge, one of the utilities of the cufflinks package, the merged assembly provides a unified basis for calculating gene expression and transcription in each expression.

2.2.2.3 Analysis differential expression with Cuffdiff

The merged assembly is sent to Cuffdiff. Adjust the Cuffdiff parameter to normalize the number of fragments mapped to a single locus to improve the strength of differential expression analysis. After Benjamini-Hochberg adjustment for multiple testing, Cuffdiff determined whether a gene was significant or not by comparing the p-value of a statistical test for differential expression against the false discovery rate (FDR). Genes with a fold change larger than 1.3 and a q-value or FDR-adjusted p-value of 0.05 were considered statistically .The overall number of DEGs, the number of ten up and down regulated genes, and the top 15 genes with differential expression changes are all evaluated in this way.

2.2.3 Pipeline 2 : HISAT, Stringtie, DESeq2.

2.2.3.1 Read alignment with HISAT

RNA sequence reads were then mapped to the human reference genome (hg38/GRCh38) by the Hisat2. Hisat2 has a parameter set to indicate alignments for transcript assemblers like StringTie. The uniformity of read coverage on exons and the mapped strand are also significant factors.

2.2.3.2 Assemble transcripts with Stringtie

Stringtie were assembled of reads aligned into potential transcripts. A novel network flow algorithm and an optional de novo assembly step to assemble full-length transcripts representing multiple splice variants for each gene location. Short read alignments or alignments of longer sequences that have assembled from those reads. Softwares such as Ballgown, Cuffdiff, or DESeq2 can be processed to identify differentially expressed genes.

2.2.3.3 Counting by Featurecounts

Once read aligned to genome the next step featureCounts can be used to get the gene counts, data input to featureCounts tool requires files of aligned reads in SAM, BAM files and a list of genomic features in either general feature format (GTF). Featurecounts tool counts reads that map to a single sites for assigning reads to a gene or an exon. This utility produces two files: a count matrix and a summary file that shows how many reads were assigned or counted, as well as why they were not.

2.2.3.4 Differential expression analysis with DESeq2

DESeq2 was performed for determining differential expression genes, DESeq2 use the gene-specific shrinkage estimation for dispersions. While assessing the dispersions, DESeq2 uses all genes with a similar average expression. The estimation of the change in folding is also used to try not to identify genes with small mean expression levels. Significantly and differentially expressed genes were statistically analyzed for the mean difference in fragment number per kilobase of copy per million reads. Conditions for finding differentially expressed genes between two groups were set with $p\text{-value} < 0.05$ and $\text{foldchange} > 1.3$. Thereby the results are analyzed including, total number of DEGs, number of ten up and down regulated genes and top 10 genes up and down differential expression changes as the same as pipeline 1

2.2.4 TopGO enrichment Analysis

Significantly and differentially expressed genes data in each pipeline were submitted to topGO to Gene Ontology Analysis (GO) for functional annotation and identification of significant biological pathways. TopGO pathways have been enriched with the aid of using R package deal to higher recognize the capabilities of the DEGs. The expected capability genes have been incorporated with the differentially expressed mRNAs within the sequencing. TopGO phrases have been assigned to decide the capabilities and pathways of the differentially expressed mRNAs. Mapping differentially expressed mRNAs to topGO phrases with inside the database after which calculated gene numbers for every time period The threshold of importance become described with the aid of using the P-value ($P < 0.05$ become taken into consideration statistically significant)

CHAPTER 3. RESULTS

3.1 Quality Control

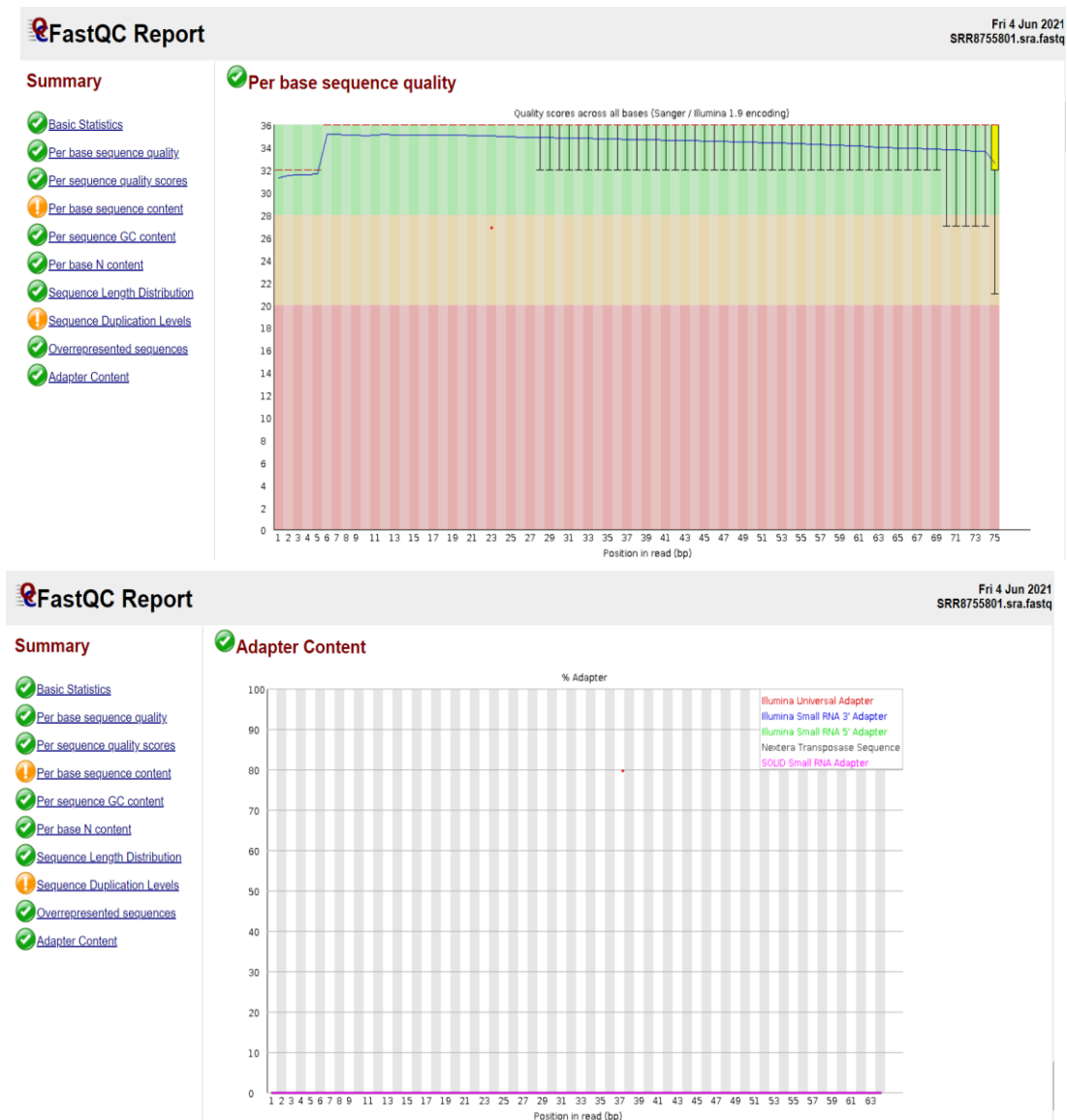


Figure 3 : Quality control results of raw reads.

FastQC is used to assess the quality of raw readings from the sequencing results before they are analyzed. The results reveal that there are no residual adapter sequences and the Q values are both above 20. Therefore, these data will be used for the next steps without going through the adapter removal process or bad quality sequence.

3.2 Alignments

Table 1 : The percent alignment ratio of two pipelines.

Name	Pipeline 1 percent read mapping rate	Pipeline 2 percent read mapping rate
SRR8755796	96,4%	97,43%
SRR8755797	96,7%	97,63%
SRR8755798	96,6%	97,56%
SRR8755799	96,6%	97,57%
SRR8755800	96,6%	97,57%
SRR8755801	96,4%	97,45%

After alignment, the overall alignment ratio showed that the alignment performance of both pipelines is at a high level. The proportion of mapped reads, which is a global measure of overall sequencing accuracy and the concentration of contaminating DNA, is an essential mapping quality statistic. Normally, the mapping rate needs to be more than 70%. In addition, comparing the two pipelines are very comparable. The research findings comparing the two pathways are quite similar between two pathways, there is a 1% variation.

3.3 Gene expression profiling

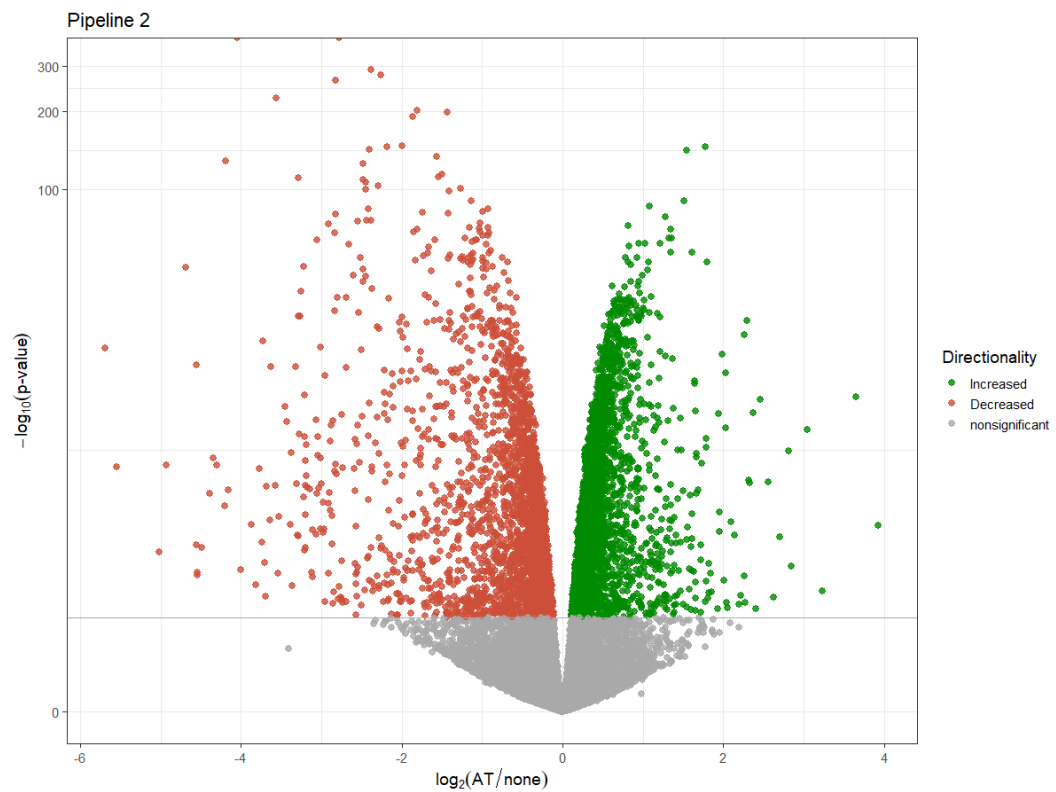
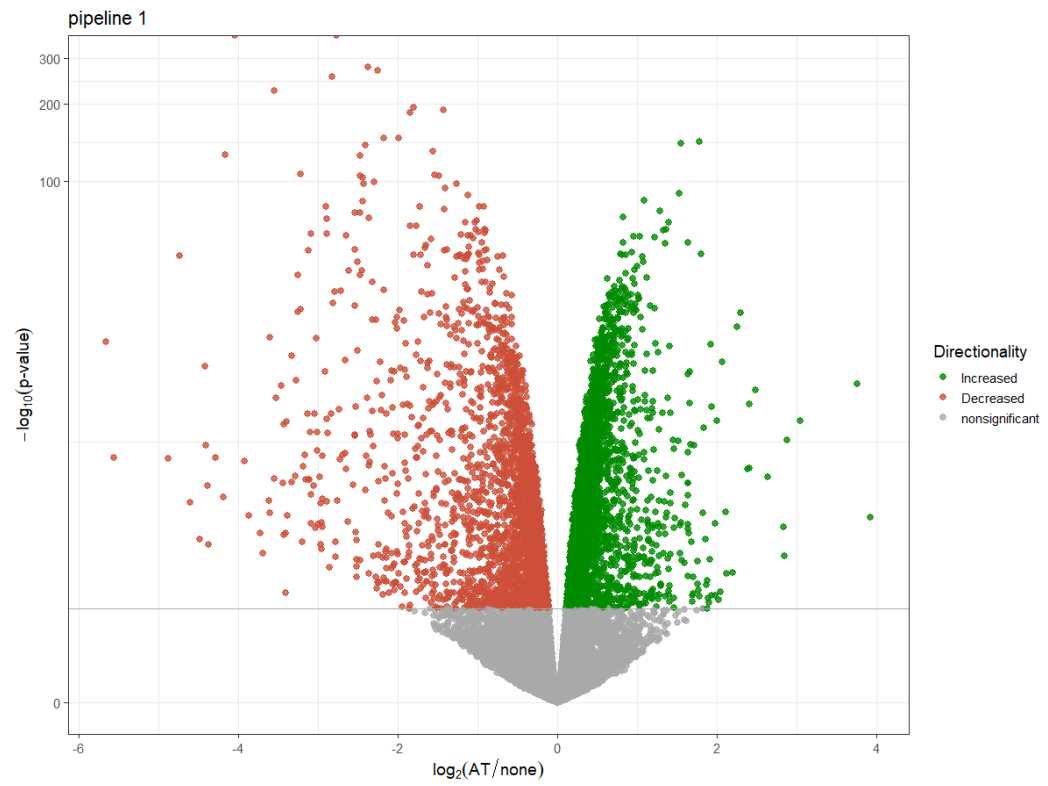


Figure 4 : Differentially expressed genes between the two samples analyzing by two pipelines.

After analysis was found that when treating with AT, the analysis in both pipelines showed that there were a large number of overexpressed genes was blue dots and genes reduce expression in the red dots ,common compare between two conditions is $-\log p\text{-value} > 0.05$ and $\text{foldchange} > 1.3$ was overexpression genes or $\text{foldchange} < 1.3$ was genes reduce expression has shown figure 3.

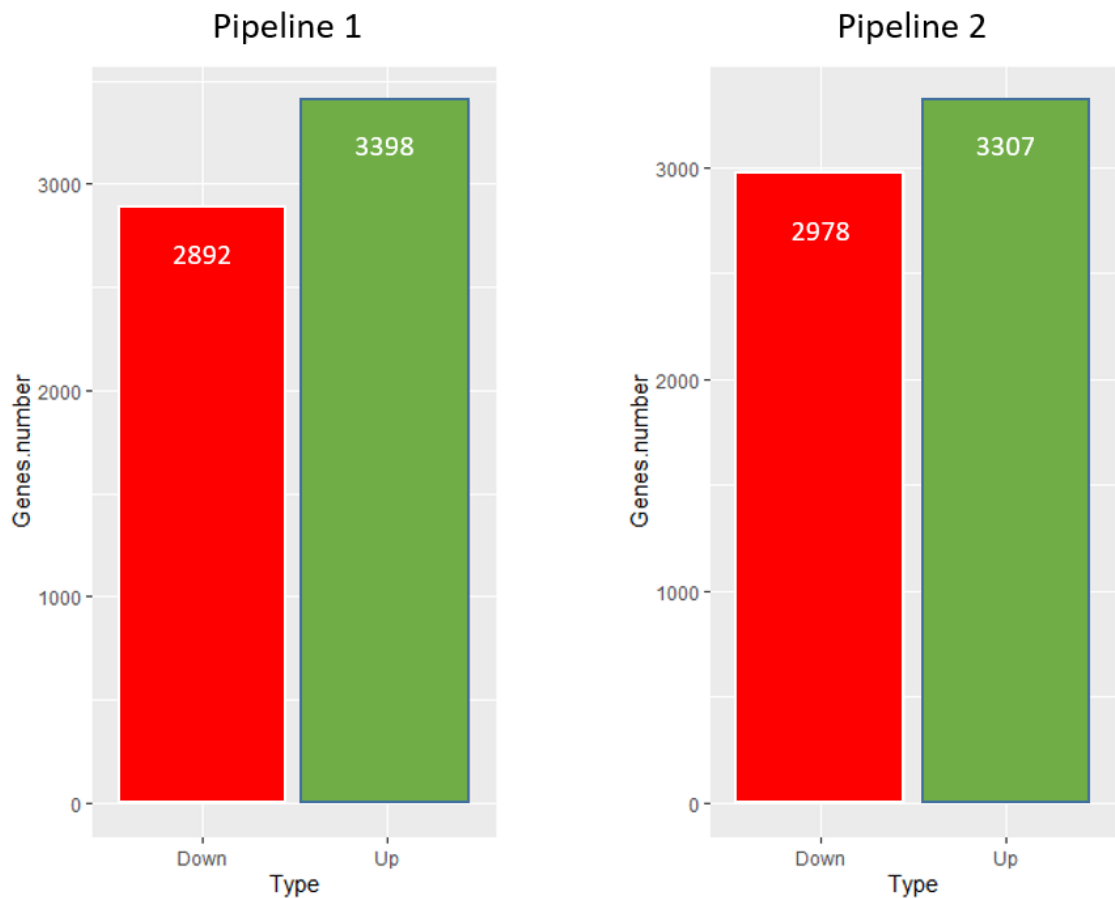


Figure 5: Bar plot of number of genes up and down regulation.

The number of genes with decreased expression in pipeline 1 was 2892 and genes with increased expression of 3398 genes, the number of genes with increased and decreased expression in pipeline 2 was 3307 and 2978 genes respectively.

Table 2: Top 10 significant genes up and down of different expression

Up Pipeline1	Log2 Foldchange	Up Pipeline 2	Log2 Foldchange	Down Pipeline1	Log2 Foldchange	Down Pipeline2	Log2 Foldchange
DRD5P2	5.02	CYP4F22	4.44	SLCO2B1	-5.65	SLCO2B1	-5.68
LPAR3	4.81	PDPN	4.31	CD86	-5.55	CD86	-5.54
PDPN	4.32	FAT4	4.23	CALHM3	-5.27	CALHM3	-5.33
CYP4F22	4.30	RP1-138B7.7	4.22	LINC00942	-5.14	LINC00942	-5.21
FAT4	4.23	AL355480.3	4.12	AC015922.5	-5.06	ABCC3	-5.21
RP1-138B7.7	4.23	KIAA1211	3.99	KRT86	-4.96	KRT86	-5.01
KIAA1211	4.12	GUCY2F	3.92	EGF	-4.88	AC015922.5	-5.00
AL355480.3	4.01	CTD-2616J11.14	3.87	PRICKLE2-AS1	-4.74	EGF	-4.92
GUCY2F	3.92	LPAR3	3.84	ANGPT4	-4.73	PRICKLE2-AS1	-4.75
CTD-2616J11.14	3.87	HTR1B	3.81	FTH1P20	-4.60	ANGPT4	-4.69

The top 10 most significant genes were sorted by log fold change (logFC). Among the top 10 genes, the two pipelines share the same gene that increases the expression LPAR3, PDPN, CYP4F22, FAT4, RP1-138B7.7, KIAA1211, AL355480.3, GUCY2F, CTD-2616J11.14. Two different gene are DRD5P2, and HTR1B. Gene HTR1B is found in the 11th row of pipeline1. Down-expression gene are include SLCO2B1, CD86, CALHM3, LINC00942, AC015922.5, KRT86, EGF, PRICKLE2-AS1, ANGPT4, two different genes are ABCC3, and FTH1P20 which is located at 20th row in pipeline2

Cancer cells are affected by the AT compound that increases the representation of the gene, for example LPAR3 in termed of LPA receptors it is also a bioactive lipid mediator that is produced

mostly from membrane phospholipids LPA signaling is involved in cell migration and proliferation, as well as cytokine production. Selective receptor antagonists in cancer growth and resistance are important since LPA receptor expression and function are necessary for cellular effects. The transmembrane receptor, commonly known as the PDPN gene, is a functional target for oral cancer and a potential chemotherapeutic target. Oral cancer invasion and metastasis are aided by PDPN expression, which increases tumor cell movement. The significance of PDPN in the progression of cell carcinoma and how it can be used to prevent and treat oral cancer, genes with increased levels of expression lead to altered pathways involved in cancer treatment.

4 Gene Ontology Analysis

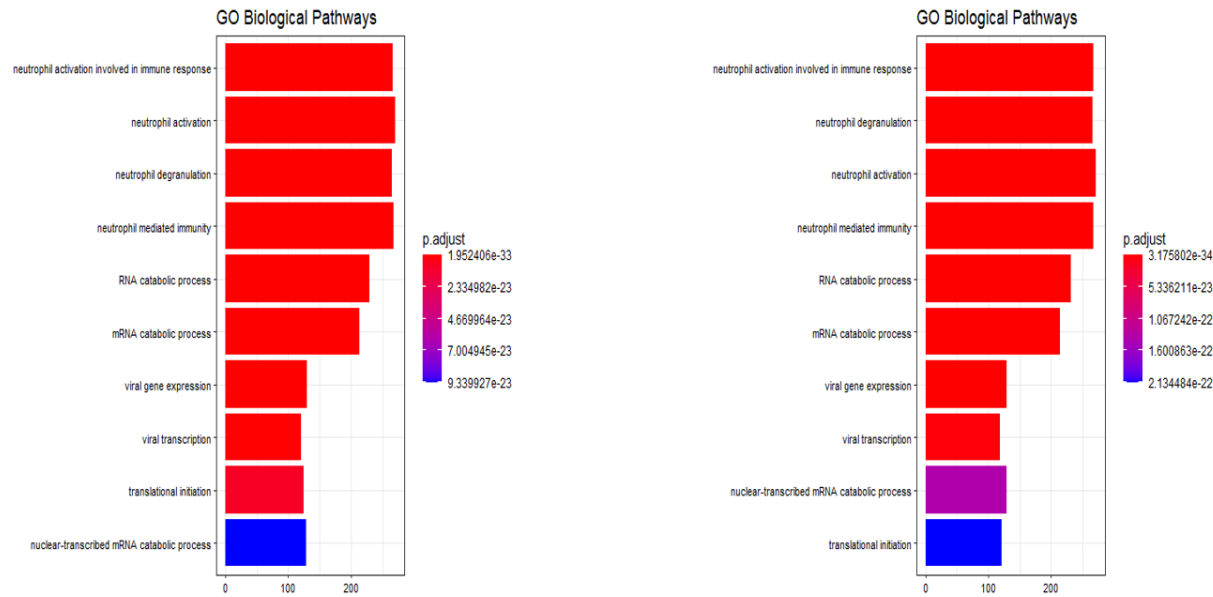


Figure 6 : TopGO enrichment of differential expression genes

After analysis, it was found that the top 10 biological pathways related to gene expression changes, when treated with AT in both pipelines, gave the same results, although the order was different. The changes were divided into three groups, immune-related (neutrophil activation involved in immune response, neutrophil activation, neutrophil degranulation, neutrophil-

mediated immunity), gene expression-related (RNA catabolic process, mRNA catabolic process), translational initiation), and the group related to viruses (viral gene expression, viral transcription)

Immunosuppressive neutrophils and anti-tumor neutrophils have several traits in common. Mac-1-mediated contact, either between neutrophil and T cell in the case of myeloid derived suppressor cell read activity or between neutrophil and antibody-opsonized tumor cell in the case of ADCC, is required for neutrophils to conduct their MDSC and anti-tumor functions. At the same time, the fact that traditional antimicrobial destruction mechanisms, such as degranulation are required for neutrophil MDSC but not for neutrophil ADCC, indicates that at least some of the required cellular mechanisms for MDSC and ADCC neutrophil activity are directly antagonistic.

CHAPTER 4. DISCUSSION & CONCLUSION

4.1. Discussion

The results of the two pipelines are similar; the difference is that pipeline 1 is typically used for machines with low RAM, while pipeline 2 can be used for machines with high RAM and large memory. However, the results are similar, so choosing a pipeline is important for each type of machine that processes data.

After treating cancer cells with the substance AT24 to kill them, researchers discovered that it altered pathways associated with immunity, gene expression, and virus infection. the genes that increases the expression participate in LPAR3, PDPN, CYP4F22, FAT4, RP1-138B7.7, KIAA1211, AL355480.3, GUCY2F, CTD-2616J11.14, DRD5P2, and HTR1B.

The goal of this research was to analyze the impacts of 4,15-iso-atriplicolide tiglate (AT), a potent inhibitor of MYB transcription, which contributes to the development of acute leukemia. Natural compound AT is a solution capable of controlling the development of leukemia and provide the potential targets for treating acute myeloid leukemia.

4.2. Conclusion

The results between the two pipelines are similar, choosing the right pipeline for the computer to process the data correctly .The study described differential expression of AT compounds among acute myeloid leukemia cells. The study expands the database of the natural compound 4,15-iso-atriplicolide tiglate and provides new potential modifiers for fecal studies on its ability to prevent and treat infectious diseases. But more research is needed to probe and confirm the expression of AT in AML cells. In addition, the detailed biological effects of the natural compounds AT remain to be elucidated in future research.

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